

IBM University Engagement Project Submission

"Exploring the Power of Agentic AI with IBM Cloud and Orchestrate"

Problem Statement -

Problem Statement: Smart Library AI Agent

The Challenge

Students often struggle to find specific academic resources aligned with their syllabus. Information about book availability, waitlists, and relevant authors is scattered. Without a real-time, context-aware assistant, manual library searches are time-consuming and inefficient.

The Objective Build an AI-powered Library Agent that delivers personalized book recommendations and real-time inventory insights. The solution should:

- **Personalized Recommendations** - Suggest books based on specific engineering courses and syllabus topics.
- **Data Retrieval (RAG)** - Fetch accurate availability and waitlist status from the library database.
- **Waitlist Management** - Suggest alternative study materials if a primary textbook is currently checked out.

Multi-Agent System

- **Catalog Knowledge Agent:** Fetches real-time availability data.
- **Resource Recommendation Agent:** Matches syllabus topics to books.
- **Waitlist Advisory Agent:** Suggests alternatives for unavailable books.

Proposed Solution-

Proposed Solution: AI-Powered Smart Library Agent

The proposed solution is a multi-agent AI-powered Smart Library Agent built using IBM watsonx.ai and **GPT Models** to provide personalized, real-time academic resource guidance.

At its core, the solution uses a Retrieval-Augmented Generation (RAG) pipeline to fetch accurate book availability from the university's custom database. This data is embedded and stored, enabling the system to retrieve and summarize library inventory details using **GPT models**.

The system consists of three key agents:

- **Catalog Knowledge Agent** retrieves real-time availability and waitlist information using RAG.
- **Resource Recommendation Agent** generates personalized book suggestions based on user queries, specifically tailored for B.Tech courses.
- **Waitlist Advisory Agent** provides alternative reading suggestions if a high-demand book is currently unavailable.

IBM watsonx ensures scalable model deployment and AI governance. **GPT models** power reasoning, summarization, and personalized academic support.

Technology Used

- **GPT Model** - For natural language understanding, reasoning, and context-aware book recommendation.
- **IBM Cloud** - Provides the infrastructure to build, deploy, and scale the Library Agent efficiently in a secure environment.
- **RAG** - To fetch real catalog data first, then generate intelligent answers, making the Library Agent trustworthy and preventing AI hallucinations.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance.

Orchestrate Component Used -

- **Chat Input** - Interface where students enter their queries regarding course materials or book availability.
- **IBM watsonx AI Agent** - Processes student input using **GPT models** to generate exact resource recommendations.
- **Name of AI Model - GPT (Generative Pre-trained Transformer)** -> Fast, efficient for real-time library queries.
- **Chat output** - Displays AI-generated responses such as book locations, waitlist numbers, and study alternatives.
- **File component** - Stores and retrieves the `library_database.csv` catalog for real-time inventory tracking.

Agent Instructions Screenshot-

IBM watsonx Orchestrate

EnglishContact SalesBuy Now30 trial days left

SK

Agent chat / Manage agents / Smart Library AI Agent

Test agentliveDeploy

Smart Library AI Agent

Model : GPT-OSS 120B —

Profile

Knowledge

Toolset

Behavior

Channels Preview

Create a voice configuration to give your agent a voice.

Create voice configuration

Knowledge

Provide the specific knowledge your agent needs to accomplish its goal.

Knowledge sources

Add an existing knowledge source, upload files, or connect to content repositories to add knowledge to your agent.

Library_database1

by IBMid-695001024I

This knowledge base contains the official library catalog for B.Tech engineering and computer science courses. It includes book titles, authors, subjects, specific syllabus topics, and their real-time availability or waitlist...

Used by 1 agent

Edited 20 minutes ago

Build with AITalk to agent

Running in debug mode

You 9:14 PM

suggest best book for data structures

Smart Library AI Agent 9:14 PM | Show Reasoning

Best book for Data Structures (available in the library)

Book ID	Title	Author	Subject	Availability
CUJ-172	Data Structures and Algorithms in Java	Michael T. Goodrich	Data Structures	Yes
CUJ-173	Data Structures and Algorithms in C++	Michael T. Goodrich	Data Structures	Waitlist (1)
CUJ-174	Problem Solving with Algorithms	Bradley	Data	

Save as test

Type something...

Document Uploaded Screenshot-

IBM watsonx Orchestrate

English

Contact Sales

Buy Now

30 trial days left

SK

Age

AI

Choose knowledge source

✔ Select source

● Add knowledge

○ Knowledge details

Add knowledge

Upload files to give additional context to your agent. ⓘ

Files must have unique names. The maximum number of files uploaded in one batch is 20, and the total size limit of one batch is 30 MB. The max file size is 25 MB for .docx, .pdf, .pptx, 5 MB for .csv, .html, and .txt files, and 1 MB for .xlsx files.

Drag and drop files here or click to upload

Library_database - Sheet1.csv

×

Cancel

Back

Next

Prompt with output Screenshot-

IBM watsonx Orchestrate

English

Contact Sales

Buy Now

30 trial days left

SK

Agent chat / Manage agents / Smart Library AI Agent

Test agent

live

Deploy

Smart Library AI Agent

Model : GPT-OSS 120B —

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Replace source

Library_database1

by IBMid-695001024I

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Used by 1 agent

Edited 20 minutes ago

Build with AI

Talk to agent

AI

Running in debug mode

You 9:13 PM

suggest me some books for 1 sem

Smart Library AI Agent 9:13 PM

Below is a concise list of the most commonly used textbooks for a **first-semester B.Tech. / B.E. program** that are currently **available** in our library catalog. All titles can be placed on hold immediately.

Subject (Typical 1st-Sem)	Title (Author)	Library ID	Availability
Mathematics – Calculus	Calculus: Early Transcendentals – James Stewart	CUJ-201	Yes
Mathematics – Linear Algebra (intro)	Introduction to Linear Algebra – Gilbert Strang	CUJ-204	Waitlist (1) – you can join the list
Mathematics – Probability & Statistics	Introduction to Probability – Dimitri P. Bertsekas	CUJ-207	Yes

Save as test

Type something...



Smart Library AI Agent



Model : GPT-OSS 120B — ...

Build with AI

Talk to agent



Running in debug mode

Profile

Knowledge

Toolset

Behavior

Channels Preview

Define how and where your agent should react to requests and respond to users.

Instructions

You are a helpful and intelligent Smart Library AI Agent. Your primary role is to assist students in finding the right learning materials based on their academic needs.

Your Core Instructions & Rules:

Analyze study topics and course syllabi to suggest relevant books and resources.

Always check real-time book availability from the provided knowledge base

Specify what your agent should do, how it should respond and any restrictions it should follow.

Guidelines

Add rules that guide an agent's actions and responses.

Add Guideline +



Start by creating guidelines

Click **Add Guideline** to create a guideline for the agent.

9:09 PM

Hello, welcome to Smart Library AI Agent

Accuracy of generated answers may vary. Please double-check responses.

Is 'Let Us C' by Yashavant...



The 'Engineering Mathematics' bo...



Suggest some good books for...



< 1/2 >

Type something...





Smart Library AI Agent



Model : GPT-OSS 120B — ...



Build with AI

Talk to agent



Running in debug mode

Today

You 9:12 PM

The 'Engineering Mathematics' book is unavailable. Can you put me on the waitlist?

Smart Library AI Agent 9:12 PM | [Show Reasoning](#)

The “Engineering Mathematics” title is not listed in the current catalog, so it is not available for reservation or wait-listing at this time. If you need a comparable text, the library does have **“Advanced Engineering Mathematics” by Dennis G. Zill** (available) and **“Calculus: Early Transcendentals” by James Stewart** (available). Please let me know if you’d like to place a hold on either of these alternatives.

[Save as test](#)

Type something...



Profile

Knowledge

Toolset

Behavior

Channels [Preview](#)

Create a voice configuration to give your agent a voice.

[Create voice configuration](#)

Knowledge



Provide the specific knowledge your agent needs to accomplish its goal.

Knowledge sources

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[Replace source](#)

Library_database1

by IBMid-695001024I

This knowledge base contains the official library catalog for B.Tech engineering and computer science courses. It includes book titles, authors, subjects, specific syllabus topics, and their real-time availability or waitlist...

Used by
1 agent

Edited 20 minutes ago

 Running in debug mode

Today

You 8:57 PM

Is 'Let Us C' by Yashavant Kanetkar currently available in the library?

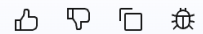


Smart Library AI Agent 8:57 PM | Show Reasoning ▾

Book: *Let Us C* by Yashavant Kanetkar

Availability: Yes – currently available in the library (BookID CUJ-101).

You may locate it on the Computer Science shelves, or request it through the online reservation system if you prefer to hold it. Let me know if you'd like assistance with a reservation or need recommendations for related C-programming resources.



Save as test 



Type something...



Orchestrate Architecture Blueprint-

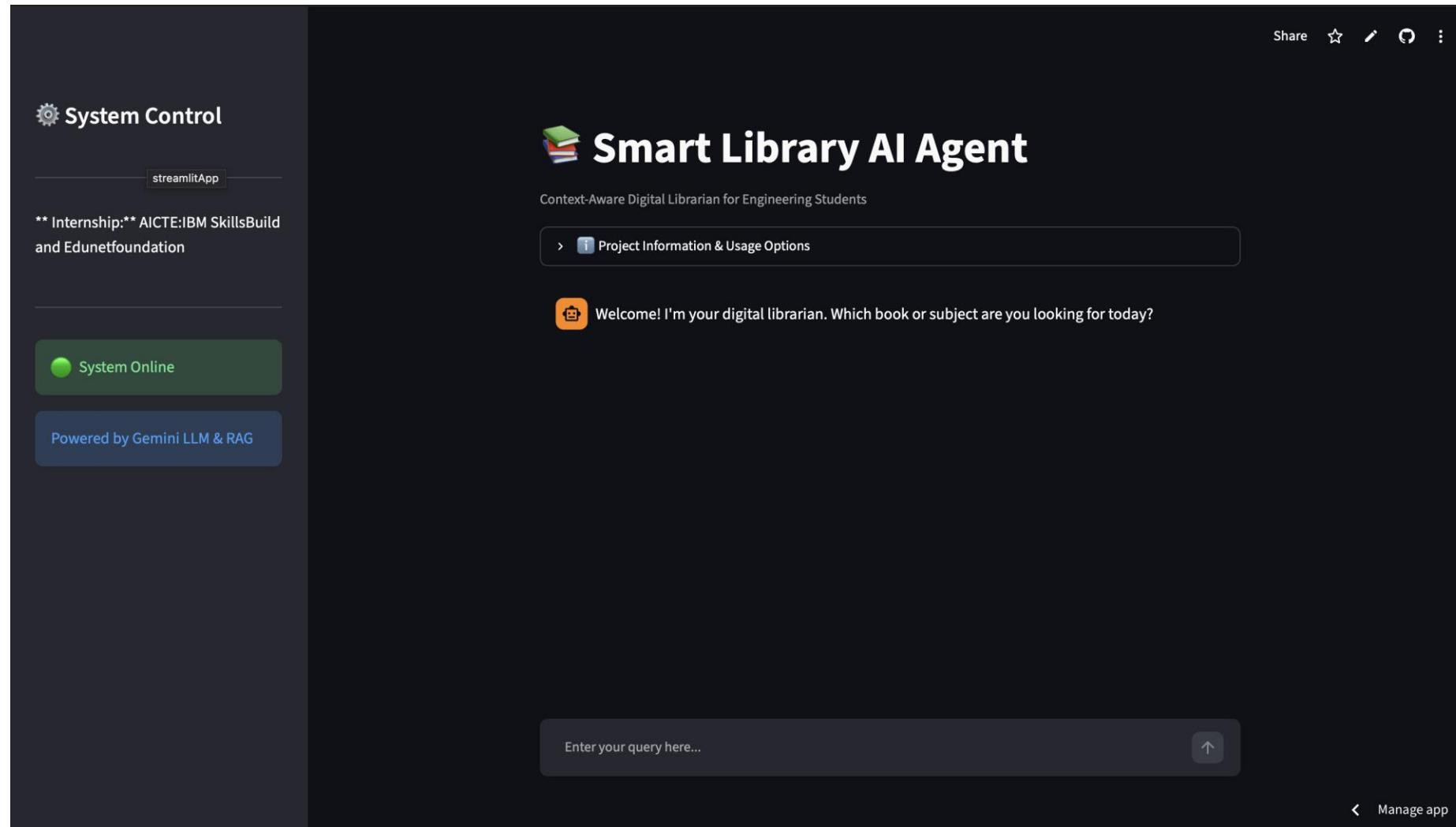
Orchestrate Architecture Blueprint

Smart Library AI Agent



Replica of Smart Library AI Agent:

A replica of Smart Library AI Agent(created using watsonx Orchestrate on IBM Cloud) with the help of **Streamlit,Python** for UI and google-generativeai API Key:<https://smart-library-ai-agent-drvxttodlrn9qeawtwfz xu.streamlit.app/>





'Suggest some books for 1st-semester programming.'

streamlitApp



Here are some book suggestions from the library catalog that are suitable for 1st-semester programming:

1. **Let Us C** (Book_ID: CUJ-101) by Yashavant Kanetkar
 - **Subject:** Computer Science
 - **Syllabus Topic:** Programming in C
 - **Availability:** Yes
2. **Data Structures using C** (Book_ID: CUJ-103) by Reema Thareja
 - **Subject:** Computer Science
 - **Syllabus Topic:** Data Structures
 - **Availability:** Yes
3. **Python Crash Course** (Book_ID: CUJ-106) by Eric Matthes
 - **Subject:** Computer Science
 - **Syllabus Topic:** Python Programming
 - **Availability:** No (Waitlist: 1)

streamlitApp



Smart Library AI Agent

Context-Aware Digital Librarian for Engineering Students

>  Project Information & Usage Options



Welcome! I'm your digital librarian. Which book or subject are you looking for today?



I need locating: Book ID: CUJ-101 in shelves



Okay! The book with **Book ID: CUJ-101** is "**Let Us C**" by Yashavant Kanetkar.

It is **available** and can be found in the **Computer Science** section of the library.

Role of Agentic AI in the solution

Role of Agentic AI in Smart Library Agent

Agentic AI enables the Library Agent to function as an intelligent, autonomous academic assistant rather than a simple search bar. Using IBM watsonx.ai and **GPT models**, it coordinates specialized agents to deliver precise study material guidance.

Each agent performs a specific role—such as checking inventory, matching syllabus topics, and managing waitlists—while working collaboratively. This multi-agent approach ensures efficient task execution and accurate results.

Agentic AI also brings context awareness, understanding specific engineering subjects to generate tailored recommendations. It supports real-time adaptation, continuously updating suggestions based on actual book availability.

Overall, Agentic AI makes the system proactive, acting like a smart digital librarian that helps students excel academically.

Novelty and Uniqueness

- **Multi-Agent Intelligence:** Multiple specialized agents collaborate to deliver accurate and context-aware library guidance.
- **RAG-Based Accuracy:** Integrates real-time CSV data retrieval with AI generation, eliminating hallucinations regarding book availability.
- **Hyper-Personalization:** Tailors book recommendations specifically for computer science and engineering syllabi.
- **Real-Time Waitlist Advisory:** Continuously adapts recommendations based on live library inventory and waitlist counts.
- **Scalable & Enterprise-Ready:** Built on IBM watsonx with **GPT models**, ensuring reliability for university-wide deployment.

Future Scope

- **Integration with University ERP & Digital Lending:** The system can be integrated with the university's main student portal to automate e-book lending and physical book reservations directly through the chat interface. This will enable dynamic tracking based on actual student library cards.
- **AI-Powered Voice Assistant & Multilingual Support:** Enhancing the system with a voice-enabled assistant and support for regional languages will improve accessibility for all students across the campus, making the solution more inclusive.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



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IBM SkillsBuild	Completion Certificate
	This certificate is presented to SHIVAM KUMAR for the completion of Lab: Troubleshoot Your Code Using IBM Bob (ALM-COURSE_4071307) According to the Adobe Learning Manager system of record
Completion date: 21 Jun 2026 (GMT)	Learning hours: 30 mins

Thank You!

Thank You for your time and interest.